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Multi-method estimation of groundwater recharge in the Fetam catchment, Upper Blue Nile Basin, Ethiopia

Mekonen Gedewon Aynalem^{1*}, Sileshi Mamo², Melkamu Meseret², Muralitharan Jothimani^{3*} and Habtamu Wondem⁴

*Correspondence:
mekonengedion07@gmail.com;
muralitharangeo@gmail.com

¹ Department of Geology,
College of Natural Sciences,
Injibara University, Injibara,
Ethiopia

² School of Earth Sciences,
Bahirdar University, Bahirdar,
Ethiopia

³ Department of Geology,
College of Natural
and Computational Sciences,
Arba Minch University, Arba
Minch, Ethiopia

⁴ Department of Geology,
College of Natural
and Computational Sciences,
Debank University, Debank,
Ethiopia

Abstract

Reliable groundwater resources are essential in catchments characterized by strong rainfall seasonality and intermittent surface-water availability, such as the Fetam catchment in the Upper Blue Nile Basin, Ethiopia. In this study, groundwater recharge was quantified using a multi-method approach to reduce method-dependent uncertainty and provide a defensible recharge range for sustainable groundwater management. Five independent methods were applied: soil moisture balance, baseflow separation of streamflow, chloride mass balance, water-table fluctuation, and catchment water balance. Hydro-meteorological data, groundwater-level measurements, hydro-chemical analyses, land use, soil, and hydrogeological information were integrated and spatially aggregated using Thiessen polygon weighting. Mean annual recharge estimates varied from 112.7 ± 2.2 to 244.9 ± 12.4 mm/year, corresponding to approximately 6.5–14% of mean annual rainfall. Higher recharge values were obtained from baseflow separation and water-table fluctuation methods, reflecting strong seasonal groundwater responses. In contrast, soil-moisture balance and water-balance methods yielded lower estimates because they explicitly accounted for evapotranspiration and runoff losses. The chloride mass balance method produced intermediate values and represents longer-term effective recharge. Recharge was found to be highly seasonal and spatially heterogeneous, with highland areas acting as major recharge zones and lowland areas functioning mainly as discharge zones. Overall, convergence among multiple methods suggests a realistic recharge range of approximately 165–220.1 mm/year for the Fetam catchment. The study demonstrates that multi-method recharge assessment is essential in data-limited, seasonally dry basins and provides a scientific basis for sustainable groundwater development and drought-resilient water management.

Keywords: Groundwater recharge, Fetam catchment, Upper Blue Nile Basin, Soil moisture balance, Chloride mass balance, Water-table fluctuation



1 Introduction

Water is essential for human health, livelihoods, and economic development, yet reliable water supplies remain challenging in regions with seasonal rainfall and intermittent surface-water flow [28]. Groundwater is therefore a dependable source for domestic use and dry-season water availability [32]. Sustainable groundwater development depends on understanding recharge, which controls the amount of water that can be abstracted without long-term declines in groundwater levels, spring discharge, or river baseflow [61, 67]. Recharge is dynamic and influenced by climate variability, land use, hydrologic connectivity, and episodic rainfall, with climate change potentially altering recharge and groundwater storage [26, 36, 45, 57]. Recharge also controls groundwater–surface-water interactions and can be modified by human interventions such as artificial recharge [24, 61, 67]. Recent evidence further shows that recharge can change rapidly following wetting events and climatic anomalies, emphasizing the need to understand its spatial and temporal variability [39].

The Fetam catchment in the Upper Blue Nile system has strong hydroclimatic seasonality, with most rainfall concentrated in a short wet season and many tributaries becoming intermittent or ephemeral during the prolonged dry season [4, 34, 40]. Under these conditions, groundwater is strategically important for sustaining dry-season baseflow, springs, shallow wells, and water supply during droughts [49, 52, 64]. However, without a quantified and uncertainty-bounded recharge baseline, groundwater development cannot be reliably assessed against replenishment, limiting yield assessment, well-field planning, drought management, and protection of groundwater-dependent ecosystems [52].

Recharge is a non-stationary flux controlled by rainfall, evapotranspiration, land use, soil properties, aquifer lithology, topography, and stream–aquifer connectivity [16, 36, 46, 52]. These interacting controls create strong spatial and temporal variability, making single-method estimates less reliable and supporting the use of multiple complementary approaches [8, 49, 52]. Ethiopian studies show substantial differences in recharge magnitude and seasonality across hydrogeological settings, including the Sana River Catchment using WetSpss [13], Wabe River Catchment using a GIS-based water balance model [56], and Maze–Zenti Catchment using WetSpss-M [43]. Method selection also affects recharge estimates, as demonstrated by soil-moisture balance, baseflow separation, WetSpss-M, and chloride mass balance approaches [6, 59, 62]. Multi-method studies further show that agreement among independent approaches provides greater confidence than reliance on a single estimator [48].

The overall objective of this research is to quantify groundwater recharge in the Fetam catchment and provide a defensible recharge range that can guide sustainable groundwater utilization. Specifically, the study aims to (1) estimate long-term recharge using chloride mass balance, (2) quantify recharge from groundwater-level variations using the water-table fluctuation method, (3) compute recharge using soil-moisture balance analysis, (4) estimate recharge using baseflow separation of streamflow records, and (5) apply a catchment water-budget approach and synthesize results across methods to characterize uncertainty and identify consistent recharge estimates for management applications.

The catchment is characterized by strong rainfall seasonality and intermittent tributary flows, which reduce the reliability of surface-water sources and elevate groundwater

as a primary dry-season supply. In such hydroclimatic settings, groundwater functions as a strategic buffer against drought; however, without quantified recharge, abstraction cannot be evaluated against replenishment, and any expansion of groundwater development risks progressive water-level decline, reduced spring discharge, and diminished dry-season baseflow that supports riverine ecosystems and downstream users [34, 52].

Recharge estimation is necessary because recharge is spatially heterogeneous and temporally variable, with its magnitude governed by interacting climatic, land-surface, and subsurface controls. These characteristics limit the transferability of recharge values between basins and mean that extrapolating from other Ethiopian catchments—or relying on generalized national estimates—can be unreliable for local groundwater management and planning [16, 18, 21, 52]. Recent studies in Ethiopia demonstrate that recharge rates and seasonal dynamics vary considerably across basins and hydrogeologic environments, and that different methods—including GIS-based distributed water balance models and WetSpss/WetSpss-M applications—may yield different recharge estimates even within the same region [6, 13, 43, 56, 59, 62], underscoring the need for catchment-specific assessment and uncertainty evaluation. Moreover, groundwater planning in a data-limited basin is particularly vulnerable to method-dependent bias; a single-method estimate can produce misleading results, whereas a multi-method framework improves robustness by constraining recharge using independent lines of evidence from the unsaturated zone, saturated zone, and streamflow response [25, 48, 52].

2 Materials and methods

2.1 Description of the study area

The Fetam River catchment, as shown in Fig. 1, is located within the Awi and West Gojam administrative zones of the Amhara National Regional State in the Upper Blue Nile River Basin, northwest Ethiopia. The catchment covers approximately 793.8 km². The river originates from the Sekela Ridge, situated to the north of the Blue Nile source. The Fetam River flows southward for about 70 km before joining the Blue Nile River at the border between the Amhara and Oromia Regional States. Both the Abay (Blue Nile) River and the Fetam River originate near a locality called Gesh Abay, diverge in direction, and later converge at Boko Tabo in Ethiopia. The catchment includes numerous perennial and ephemeral streams, as well as springs, which provide continuous contributions to the Fetam River. The total population within the catchment, based on data collected from each district, exceeds 250,000, with a population density of approximately 504.8 people per km². The majority of the inhabitants, around 80% of the labor force, are engaged in agricultural activities. Geographically, the catchment is bordered on the east by the Birr River catchment, which originates from the Choke Shield and nearby mountains, including Satma, Mizan, and Jambera. The Blue Nile Gorge encircles the area from the north to the south. The primary water source appears to be the Choke Shield, where deep subsurface water circulation forms depression springs at Gish Abay and flows northward, as shown by the pink-bordered area in Fig. 1. The Fetam River is genetically related to the Birr, Blue Nile, and Zingini Rivers, with the Zingini River located on the western side of the Fetam catchment.

The catchment supports a population exceeding 250,000, with a population density of approximately 504.8 people km², and about 80% of the labor force is engaged

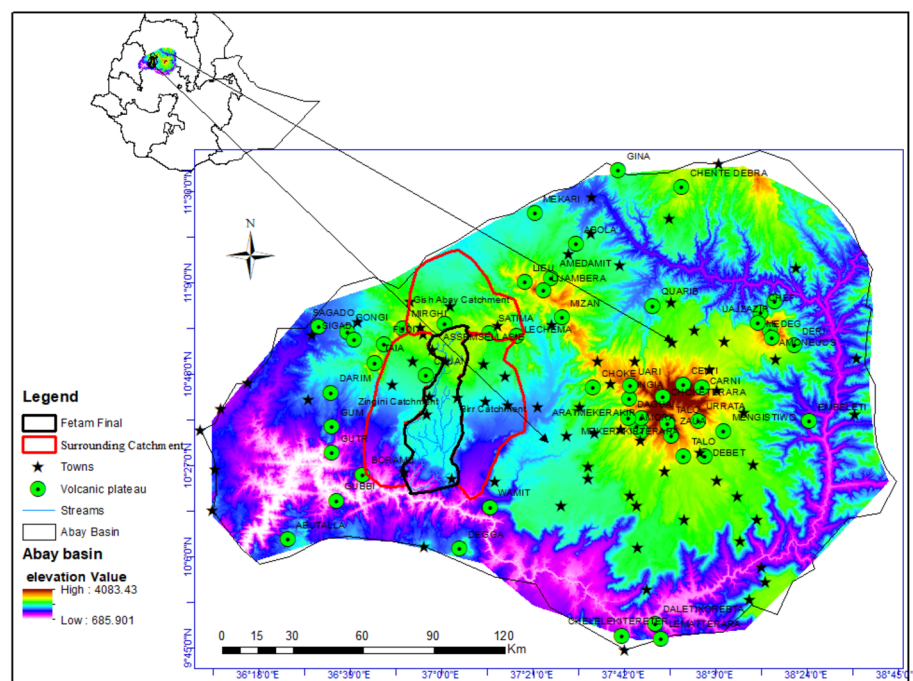


Fig. 1 Location map of the Fetam catchment

in agriculture, reflecting the strong dependence of local livelihoods on land and water resources [40]. The catchment is bordered to the east by the Birr River catchment, originating from the Choke Shield and adjacent highlands, while the Blue Nile Gorge bounds parts of the area from north to south. Climatically, Ethiopia is influenced by the seasonal migration of the Intertropical Convergence Zone (ITCZ) combined with pronounced altitudinal contrasts, which strongly control spatial and temporal climate variability across the country [40]. The study area spans the commonly recognized Ethiopian agro-climatic belts, including the Dega (temperate) and Winedega (subtropical) belts.

2.2 Elevation, soil, and lithology

Topographically, the catchment is characterized by relatively gentle relief, with slopes of 0–5% covering about 70% of the basin, 5–10% covering approximately 10%, and the remaining ~20% mainly within the 10–30% range, indicating that much of the area is flat to gently undulating. Land use and land cover are dominated by cultivated land, bushland, forest, grassland, and rural and urban settlements, consistent with patterns reported for Ethiopian agricultural catchments [55]. Cereal crops and grasses together account for more than 85% of the catchment area, underscoring the dominance of rain-fed agriculture. Major crops grown in the area include teff, shimbira, maize, potato, and onion, which are widely cultivated across Ethiopian highlands and contribute significantly to national food production [7, 30, 60].

Clay-rich units dominate the soils of the catchment, with Eutric Vertisols and Humic Nitisols covering more than 95% of the area, while Chromic Luvisols and Dystric Lep-tosols occupy smaller proportions. Eutric Vertisols are widely distributed on plains and plateaus and are characterized by a clayey texture and imperfect drainage, which

strongly influence infiltration and runoff processes [41, 42]. In contrast, Humic Nitisols are generally well-drained, with clay to silty clay textures and strongly acidic conditions, making them favorable for deep rooting and the development of subsurface flow [42]. Dystric Leptosols mainly occur on hilly and steep escarpments, where shallow soil depth and active erosion processes dominate. At the same time, Chromic Luvisols are typically found on high plateaus and open plains with moderate drainage characteristics [41]. The study area features a dense, mature stream network with a predominantly dendritic drainage pattern. The Fetam River, a tributary of the Blue Nile, and its associated tributaries originate on the northern plateau and flow toward the southwestern escarpment, reflecting topographic control on drainage development and characteristic dendritic network formation in structurally homogeneous terrains [17].

The geology of the study area comprises a sequence of Mesozoic sedimentary rocks, extensive Tertiary volcanic formations, and Quaternary volcanic and unconsolidated deposits (Fig. 2). At the southern margin of the catchment, Mesozoic sandstone crops

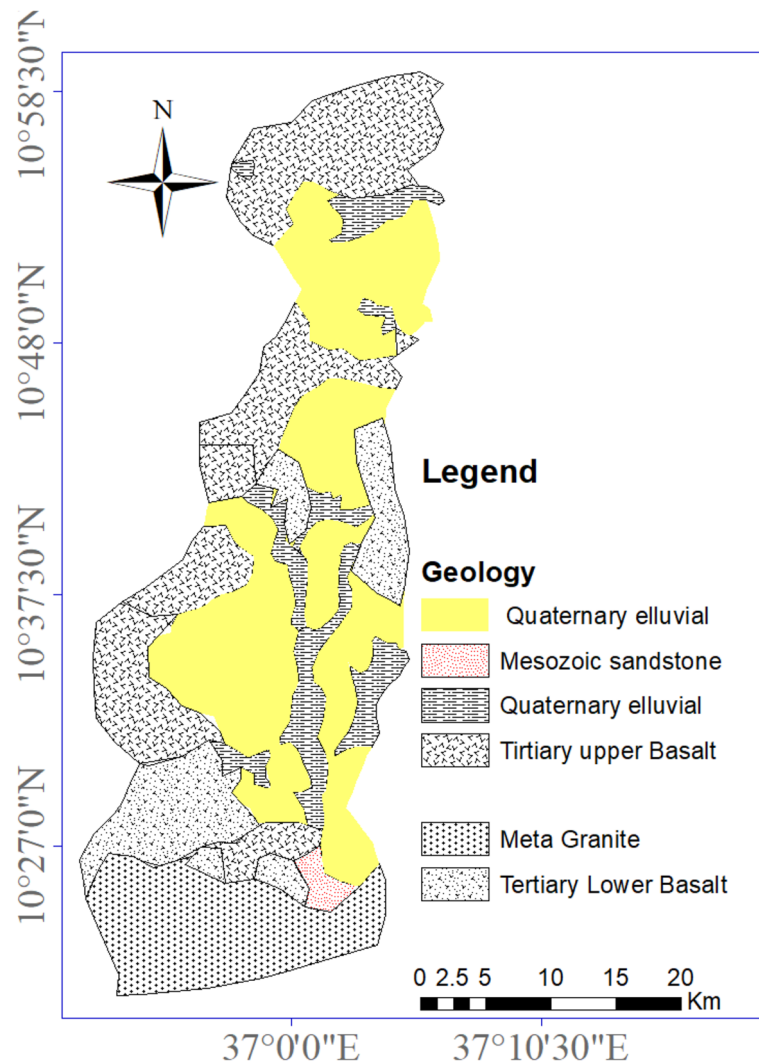


Fig. 2 Geological map of the Fetam catchment

out and is unconformably overlain by Tertiary volcanics; it is described as friable, commonly jointed, reddish near its contacts, and attains thicknesses of up to about 957 m. The volcanic succession includes Tertiary lower basalts, representing the basal trap series unit, which are deeply weathered and irregularly jointed with thicknesses of about 700 m, and Tertiary upper basalts that occur conformably above the lower basalts and include tuff intercalations and scoriaceous flows, forming cliffs and reaching about 600 m thickness in places. Quaternary volcanics arise mainly in the north-central catchment as scoria cones and crater-rim bodies composed of scoriaceous basalt pyroclasts, and as the youngest thin scoriaceous basalt flows, exposed along river cuts around Injibara and Sekela, typically 5–7 m thick, with well-developed columnar/vertical joints and minimal soil development. Unconsolidated Quaternary eluvial deposits widely mantle plateau surfaces as red to reddish-brown silty to sandy materials with basalt fragments (up to about 5 m thick) interpreted as in-situ weathering products, while Quaternary alluvial deposits occur along river valleys and marshy areas—especially near the main river and around Injibara and Sekela—commonly as black cotton soils and locally dark brown silty sediments that blanket the underlying bedrock.

2.3 Data used

This study integrated hydro-meteorological, hydrological, groundwater, hydrochemical, and catchment-parameter datasets to estimate groundwater recharge using multiple methods. Meteorological data from stations within and around the catchment included rainfall, temperature, wind speed, relative humidity, sunshine hours, evaporation, pressure, and radiation, with reference evapotranspiration calculated using CROPWAT. Approximately 15 years of streamflow data were used for baseflow separation, while runoff from ungauged areas was estimated using a runoff coefficient of 0.5. Monthly groundwater levels from about 15 hand-dug wells were used for the WTF method, with specific yield values adopted from the literature. Chloride concentrations in rainfall, groundwater, and surface water were analyzed for the CMB method. Soil, vegetation, root depth, and plant-available water parameters were also compiled to support soil-moisture-balance recharge estimation.

2.4 Methods (Groundwater recharge estimation)

2.4.1 Overview of the recharge-estimation framework

Groundwater recharge was quantified using multiple complementary approaches: (i) soil moisture balance (daily and monthly time steps), (ii) base flow separation (including recession-curve analysis), (iii) chloride mass balance (CMB), (iv) catchment water balance, and (v) water table fluctuation (WTF). Applying several methods in parallel reduces single-method bias and provides consistency checks across climate, streamflow, hydrochemical, and groundwater-level evidence. Hydro-meteorological inputs (precipitation, temperature, wind speed, relative humidity, and sunshine hours) were compiled as multi-year daily and monthly datasets from local meteorological stations, while multi-year daily river discharge was obtained for the main river gauge in the catchment. Point-based recharge estimates were spatially aggregated to the catchment scale using Thiessen polygon weighting.

2.4.2 Reference evapotranspiration (ET_o) and potential evapotranspiration (PET)

Reference evapotranspiration (ET_o) was computed using the FAO Penman–Monteith method implemented in CROPWAT 8.0 from station meteorological data (daily and separately monthly) [53, 68].

PET was derived from CROPWAT ET_o using a multiplicative factor Eq. 1:

$$PET = 0.88ET_o \quad (1)$$

where: PET=potential evapotranspiration [mm day^{−1}], ET_o=reference evapotranspiration from CROPWAT [mm day^{−1}], 0.880=local scaling factor applied in this study.

2.4.3 Soil moisture balance (SMB): root-zone water accounting (daily and monthly)

Recharge was estimated as deep percolation (drainage) below the root zone whenever updated soil–water storage exceeds the maximum plant-available water capacity (PAW) [12, 47, 58]).

2.4.4 Effective precipitation after runoff

Runoff was represented using a runoff coefficient approach Eqs. 2 and 3, [3].

$$Pe = P - Ro \quad (2)$$

$$Ro = CrP \quad (3)$$

where: Pe=effective precipitation entering soil after runoff [mm day^{−1}], P=precipitation [mm day^{−1}], Ro=runoff depth [mm day^{−1}], Cr=runoff coefficient [−] (dimensionless).

Different Cr values were applied for sub-areas relative to the gauging station (e.g., higher runoff coefficient below the gauge based on slope and land cover; upper-area coefficient derived from runoff/precipitation ratios), and the resulting effective precipitation was passed into the root-zone bookkeeping to compute soil storage and drainage.

2.4.5 Root-zone storage update and recharge (deep drainage)

A simplified daily (or monthly) storage update consistent with Thornthwaite–Mather [15] accounting using the following Eq. 4:

$$St = \min(PAW, St - 1 + Pe - AET_t) \quad (4)$$

Recharge (deep drainage) is then computed when storage would exceed PAW Eq. 5:

$$Rt = \max(0, St - 1 + Pe - AET_t - PAW) \quad (5)$$

where: St=root-zone soil water storage at time t [mm], PAW=maximum plant-available water in the root zone [mm] (depends on soil water-holding capacity and rooting depth) [33]. AET_t=actual evapotranspiration at time t [mm time^{−1}], Rt=recharge (deep drainage) at time t [mm time^{−1}].

The Excel-based SMB updates storage using precipitation and PET inputs, computes AET, tracks accumulated potential water loss (APWL) as part of the Thornthwaite–Mather logic, and assigns recharge whenever storage exceeds PAW [12, 58].

2.4.6 Baseflow separation method and scaling to ungauged areas

Baseflow separation interprets groundwater contribution to streamflow (baseflow) as an integrated response to recharge (Mamo 2015). Baseflow index was calculated using Eq. 6.

$$BFI = Q_b/Q \quad (6)$$

where: BFI=baseflow index [–], Q_b =baseflow discharge volume (or mean discharge) [$m^3 s^{-1}$], Q =total streamflow discharge volume (or mean discharge) [$m^3 s^{-1}$].

The study computed baseflow fractions after hydrograph separation, then used the separated baseflow series to estimate groundwater contribution and infer recharge timing/magnitude at the basin scale (Mamo 2015).

2.4.7 Chloride mass balance (CMB) method

Recharge was estimated using a chloride mass-balance formulation, which relies on chloride's conservative behavior in the hydrologic system ([63, 5]. The present study used Eq. 7 adopted from:

$$R = C_p P - C_q Q / C_{gw} \quad (7)$$

where: R =groundwater recharge [$mm\ time^{-1}$], C_p =chloride concentration in precipitation [$mg\ L^{-1}$], P =precipitation [$mm\ time^{-1}$], C_q =chloride concentration in runoff [$mg\ L^{-1}$], Q =runoff [$mm\ time^{-1}$], C_{gw} =chloride concentration in groundwater [$mg\ L^{-1}$]. Chloride was measured in rainfall, groundwater, and surface water samples across the catchment; precipitation was represented using Thiessen-weighted station rainfall, and runoff was estimated from gauged discharge and runoff coefficient estimates in ungauged areas (Garg 2005).

2.4.8 Catchment water balance method

A catchment water balance approach was used to estimate recharge, assuming that net groundwater inflow approximately equals net groundwater outflow at the basin scale [54]. The Eq. 8 applied in this study is:

$$R = P - (R_o + AET) \quad (8)$$

where: R =groundwater recharge [$mm\ time^{-1}$], P =precipitation [$mm\ time^{-1}$], R_o =runoff [$mm\ time^{-1}$], AET =actual evapotranspiration [$mm\ time^{-1}$]. AET was estimated using (i) SMB outputs and (ii) Thornthwaite–Mather logic (wet periods: $AET \approx PET$; dry periods: $AET = P + \text{soil-moisture withdrawal}$), and the final AET used in the catchment water balance was taken as an average of the two.

2.4.9 Water table fluctuation (WTF) method

The WTF method assumes that rises in groundwater level are caused by recharge and that specific yield remains approximately constant over the analysis period [21]. Groundwater head computation (from GPS elevation and measured depth to water). Equation 9 was used to measure the WTF.

$$h = z_{\text{ground}} - d \quad (9)$$

where: h = groundwater head (water level elevation) [m]. z_{ground} = ground elevation from GPS [m]. d = depth to water measured in the well [m].

Monthly monitoring across hand-dug wells provided a time series of head; the amplitude between post-rainy-season highs and dry-season lows represented recharge-related change.

2.4.10 Recharge from water-level rise (or representative amplitude)

Recharge from water-level rise was calculated using Eq. 10.

$$R = Sy\Delta h/t \quad (10)$$

where: R = groundwater recharge [mm time⁻¹], Sy = specific yield [–] (dimensionless), Δh = rise in groundwater level (representative seasonal amplitude) [m], t = time interval associated with Δh [day] or [year] [21].

2.4.11 Spatial aggregation to catchment scale (Thiessen weighting)

Station-based recharge (or rainfall/PET inputs) was upscaled to catchment scale using Thiessen polygon (Eqs. 11 to 13) weights:

$$X_{\text{catch}} = \sum_i^n w_i X_i \quad (11)$$

$$w_i = A_i / A_{\text{tot}} \quad (12)$$

$$\sum w_i = 1 \quad (13)$$

where: X_{catch} = catchment-average variable (rainfall, recharge), [mm time⁻¹], X_i = value at station i [mm time⁻¹], w_i = Thiessen weight for station i [–], A_i = polygon area represented by station i [km²], A_{tot} = total catchment area [km²].

3 Results and discussions

This section presents the outcomes of the groundwater recharge assessment and interprets them in relation to the catchment's climate, physiography, land use/land cover, and hydrogeological setting. Recharge was quantified using multiple independent approaches—including soil moisture balance (daily and monthly), water-level fluctuation, and baseflow separation—to capture both the magnitude of recharge and its spatiotemporal variability. Applying several methods is essential because recharge is not directly measurable at the catchment scale and each technique responds differently to

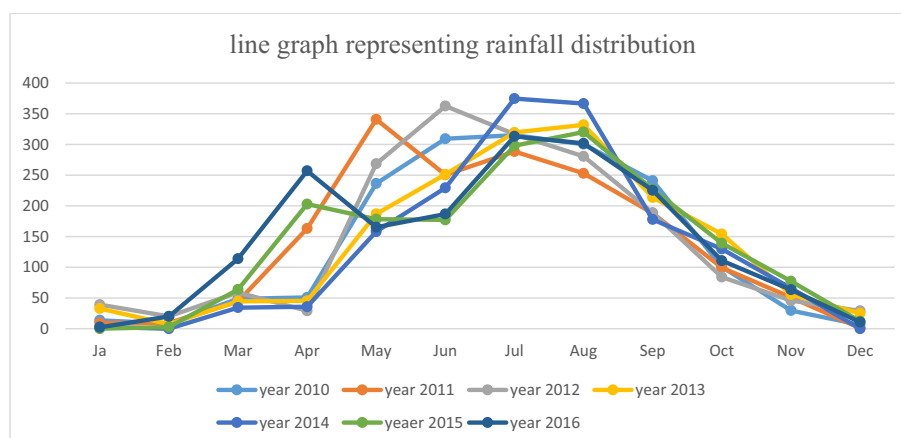


Fig. 3 Mean average rainfall distribution in the study area

Table 1 Monthly mean and standard deviation of precipitation and associated variability bands for 2010–2016

Month	Mean	SD	Lower band	Upper band	Band width
January	14.30	15.64	– 1.35	29.94	31.29
February	9.67	7.78	1.89	17.45	15.56
March	58.37	26.39	31.98	84.76	52.78
April	112.00	93.77	18.23	205.77	187.54
May	219.20	66.97	152.23	286.18	133.95
June	252.41	65.68	186.72	318.09	131.37
July	318.00	27.43	290.57	345.43	54.86
August	307.76	36.60	271.16	344.36	73.20
September	208.76	23.97	184.79	232.74	47.95
October	116.83	24.94	91.89	141.77	49.88
November	55.56	15.46	40.10	71.03	30.93
December	12.02	11.24	0.78	23.25	22.48

data resolution, storage assumptions, and the degree of groundwater–surface water connectivity [21, 46, 49, 52]. Therefore, agreement among methods increases confidence, while differences are interpreted in terms of method sensitivity and local controls.

3.1 Precipitation

Precipitation is the major component of hydrometeorology in the groundwater study. The average precipitation at three stations in the catchment is shown in Fig. 3 below. The spatial and temporal average precipitation in the area reaches its maximum during the summer season, from May–September, and its variations are shown in Table 1 below.

The spatial average precipitation at each station in Tillili, Bure, and Shendi is almost identical. The maximum rainfall at each station occurs during the summer season, from Jun to September. May and October have intermediate rainfall. The rainfall records are 1880 mm per year, 1805 mm per year, and 1400 mm per year at Tillili, Shendi, and Bure, respectively, and the daily precipitation over a period of 10 years at Tillili, 8 years at Bure, and 9 years at Shendi is 2027, 1524, and 1739 mm/year,

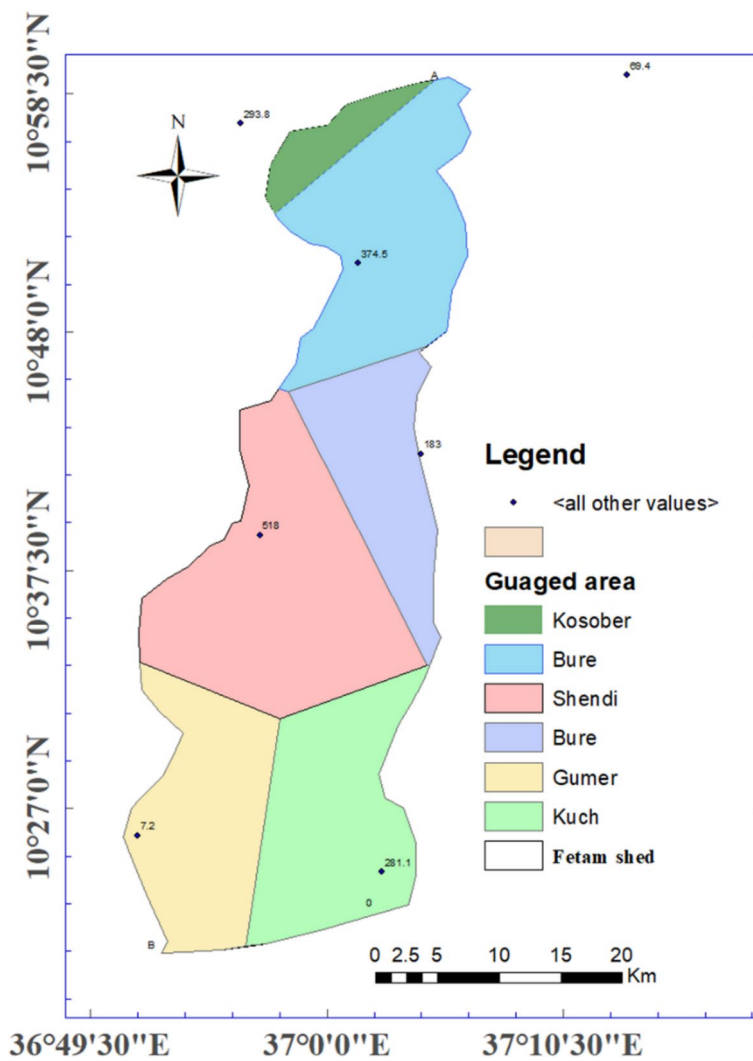


Fig. 4 Thiessen polygon

Table 2 Rainfall distribution by Thiessen polygon

No	Station	Rainfall	Area(km ²)	Weighted area%	Weighted rain
1	Sekela	1840	30.2	3.8	69.9
2	Bure	1400	103.9	13.1	183.0
3	Shendi	1805	228.3	28.8	518.0
4	Kosober	1813	128.7	16.21	293.9
5	Kuch	1602	139.4	17.6	281.3
6	Tillili	1880	158.2	19.9	374.5
7	Gumer	1115	5.3	0.7	7.4
Total		1740	793.9	100	1727.2

respectively. The average becomes 1763.3 mm/year. Precipitation measurements at each meteorological station cover only a specific location and yield an average of 1695 mm/year. The mean monthly rainfall in the catchment, calculated using the

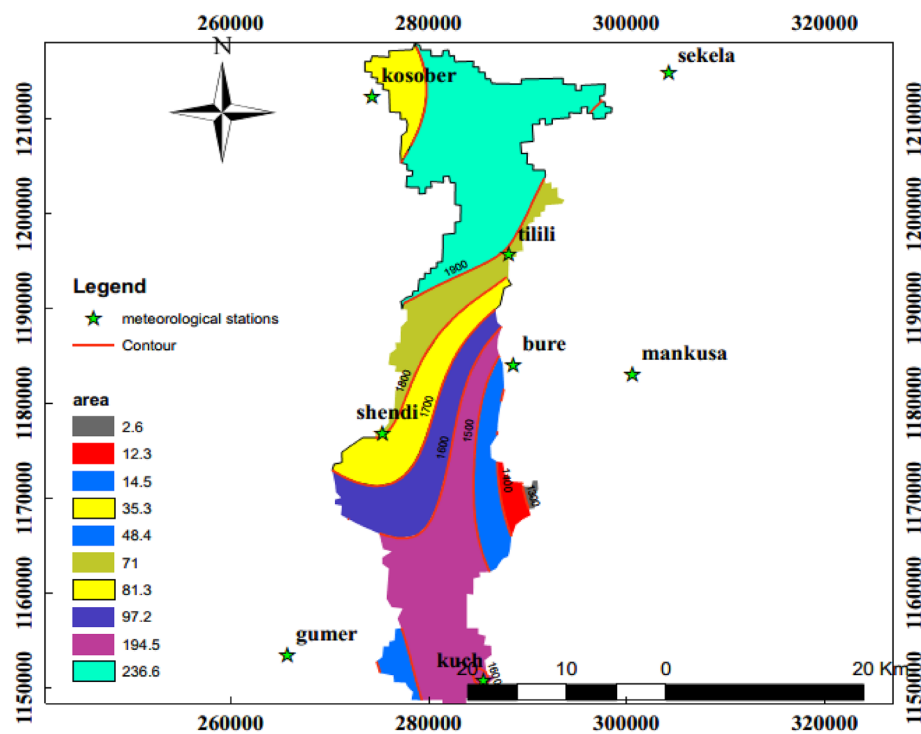


Fig. 5 Isohyetal map

Table 3 Distribution of rainfall by isohyets

Precipitation	Average value of isohyets(mm)	Area between isohyets(mm)	Weighted area(sq.-km)	Weighted R
<1300	1300	2.6	0.3	3.9
1300-1400	1350	12.3	1.55	20.25
1400-1500	1450	63	7.9	144.55
1500-1600	1550	194	24.44	378.82
1600-1700	1650	97.2	12.2	201.3
1700-1800	1750	116.6	14.7	257.3
1800-1900	1850	97.5	9.0	166.5
>1900-1900	236.6	29.81	566.4	
Total	1600	793.8	100	1709

Thiessen polygon method, is 1727 mm/year from the seven stations, and from the daily data, it is 1825.7 mm/year. Spatially weighted rainfall was computed using the Thiessen polygon method (Fig. 4). Thiessen-weighted rainfall contributions for each station are summarized in Table 2.

The total amount of weighted rainfall in the Isohyetal method is 1709 mm per year (Fig. 5). Among the applied methods, the result estimated by the arithmetic mean is lower than that estimated by the other two methods. The mean annual rainfall at each method is 1695, 1709, and 1727.2 mm/year using arithmetic isohyets and Thiessen polygons, respectively. Isohyetal rainfall-area weighting results are summarized in Table 3.

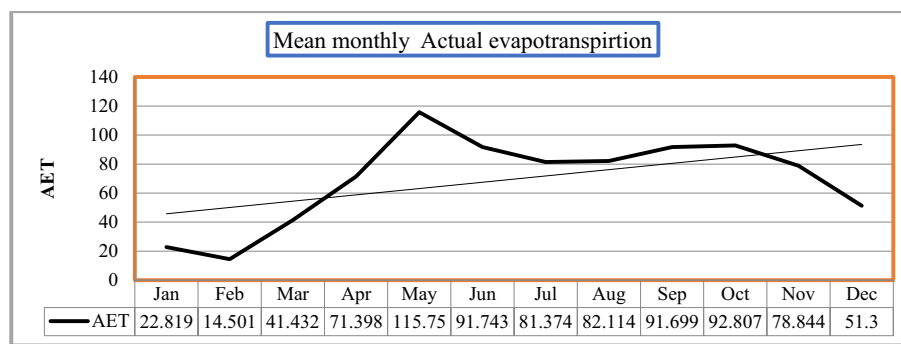


Fig. 6 Distribution of actual evapotranspiration

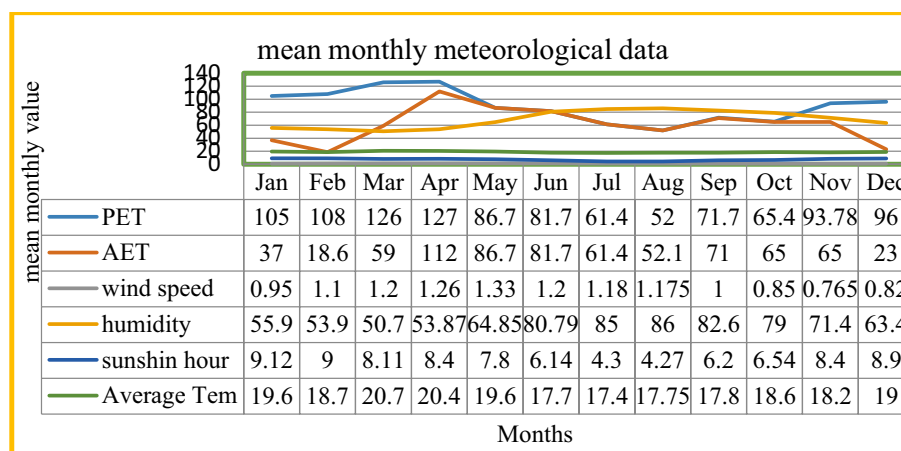


Fig. 7 Distribution of meteorological parameters

3.2 Actual evapotranspiration

Actual evapotranspiration (AET) is a key controlling factor on groundwater recharge in water-balance and soil-moisture-balance approaches because it represents the most significant “loss” term competing with infiltration and deep percolation; when AET is high, a greater fraction of rainfall is consumed by atmospheric demand and less remains available for effective recharge [46, 52]. The monthly AET pattern in Fig. 6 shows a strong seasonal cycle, with low values in January–February, a rapid rise through March–April, and a clear peak in April–May, indicating that evaporative demand and soil–plant water use intensify toward the end of the dry season as temperatures and atmospheric demand increase while soil moisture is still being depleted [12, 68].

Figure 7 illustrates that PET and AET converge during the wet season, when rainfall replenishes soil water and vegetation has sufficient moisture so that actual losses can approach the potential (energy-limited) rate; under these conditions, the partitioning of rainfall shifts toward sustaining evapotranspiration and restoring soil storage before substantial deep percolation can occur [15, 46, 68]. In contrast, during the dry season PET remains relatively high while AET declines, producing the opposite trends noted in the text, this divergence reflects moisture limitation, where atmospheric demand persists but actual evapotranspiration is constrained by depleted soil water, a dynamic that

delays effective recharge until wet-season rainfall pulses first overcome soil moisture deficits [15, 31, 46].

3.3 Land use land cover

Land use change is a major factor for water resources distribution [66].

Five major land use/land cover (LULC) classes characterize the area: built-up land, grassland, cereal cropland, forest, and bare land, as mapped in Fig. 8 using a Sentinel image processed through unsupervised classification and then refined through field verification to ensure that the spectral classes correspond to real ground conditions [55, 60]. This workflow is appropriate for heterogeneous agricultural landscapes because unsupervised clustering first captures dominant spectral groupings in the imagery, while field checks reduce class confusion that commonly occurs between spectrally similar covers such as cropped fields, dry grasslands, and sparsely vegetated bare surfaces [30, 60].

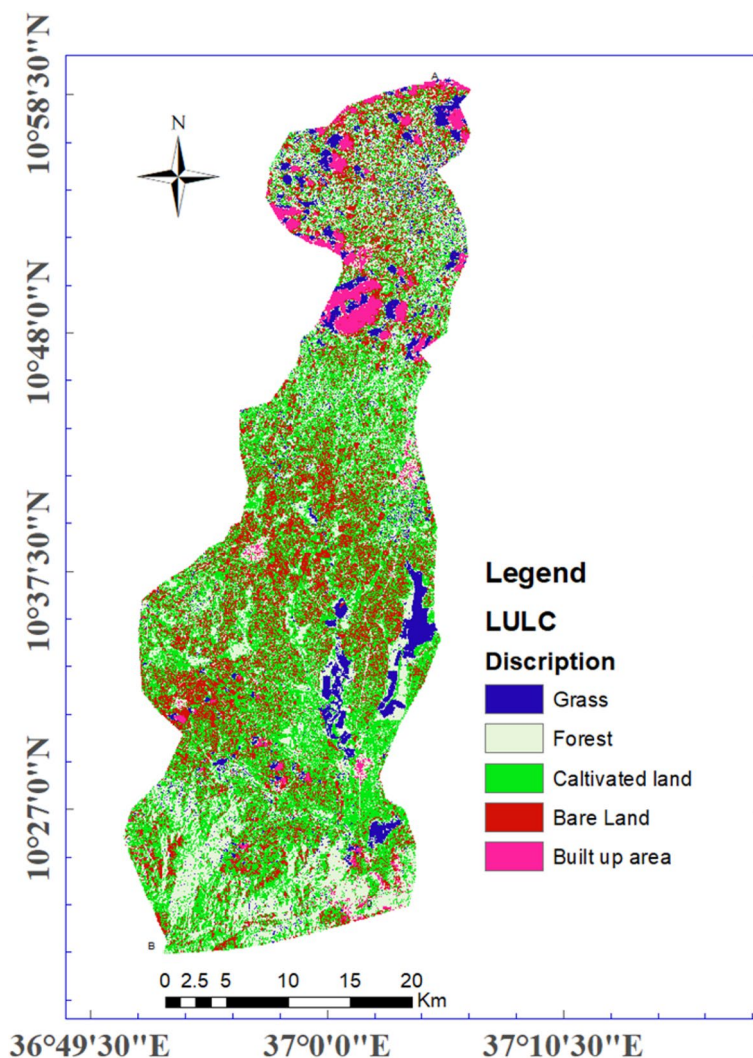


Fig. 8 LULC map of the area

The resulting map shows that the landscape is cropland-dominated, with cereal crops forming the most extensive class, followed by grasslands, indicating that rainfed farming is the primary land use and that grazing or open herbaceous cover forms the secondary matrix across the catchment [7, 60]. In contrast, forest appears more limited and spatially constrained, while bare lands likely represent areas with sparse vegetation cover, exposed soil/rock, or seasonally fallow/degraded surfaces, built-up areas occupy the smallest proportion but remain important because they concentrate human activity and can locally modify runoff generation and infiltration conditions [34, 55]. Overall, the dominance of cereal cropland and grassland suggests that land management and seasonal vegetation dynamics are likely to be key controls on water balance partitioning and, therefore, on the spatial variability of recharge across the catchment [36, 60].

3.4 Groundwater flow system

The general flow direction of subsurface water in the catchment is north to south in the northern part, following the surface water flow direction, and north-west to south-east in the southern part, perpendicular to surface water due to concealed tectonic structures. Variation in hydraulic conductivity among different rocks can also cause local deviations in groundwater flow. The contour lines shown in the figure above are parallel and spaced at uniform intervals; this indicates that the topography is nearly uniform and that groundwater flow closely follows the surface water flow system. The contours have the highest elevation in the Fetam catchment than in the Yessir catchment, and most of the borehole lithological logs indicate that the underlying rock is a fractured aquifer in

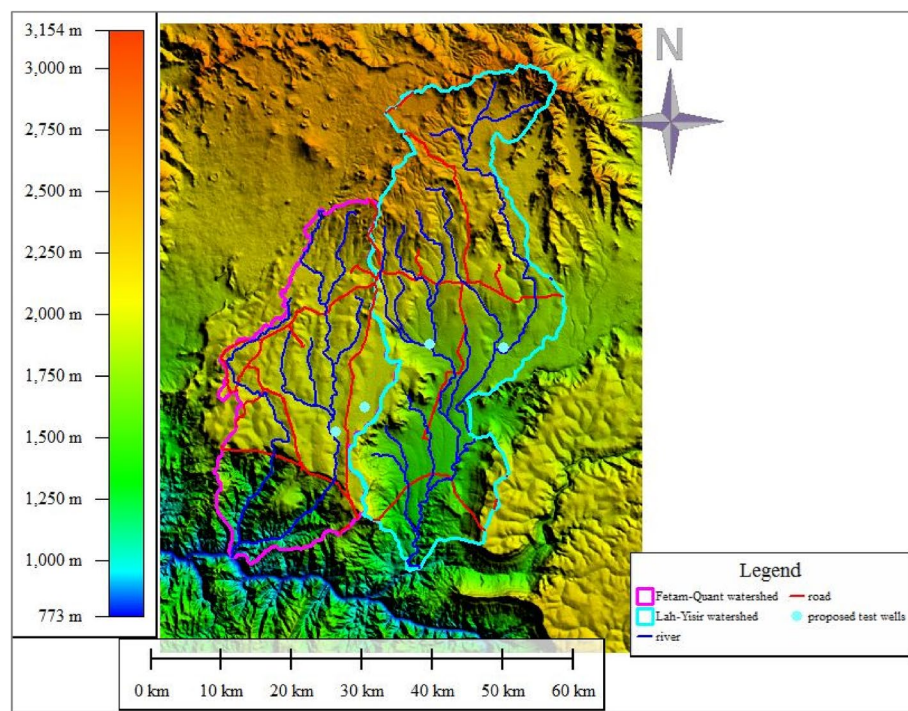


Fig. 9 Topographic divide between Fetam and Yessir

both catchments. The topographic divide between the two catchments is indicated in Fig. 9.

There is no surface water flow from the Fetam catchment to Yessir, but the groundwater flow system, integrated with the underlying fractured aquifer, flows from the Fetam catchment to Yessir. Generally, groundwater flows from recharge areas to discharge areas, from higher to lower groundwater heads. Tillili and its surrounding areas in the catchment are recharge areas. Some parts of Bure and Shendi are discharge areas; the rest of Tillili, Bure, and Shendi are intermediate zones. The inferred groundwater flow direction is illustrated in Fig. 10.

3.5 Conceptual model framework

As illustrated in Fig. 11, the conceptual hydrogeological model of the Fetam Catchment integrates the interconnected recharge, evapotranspiration, groundwater flow,

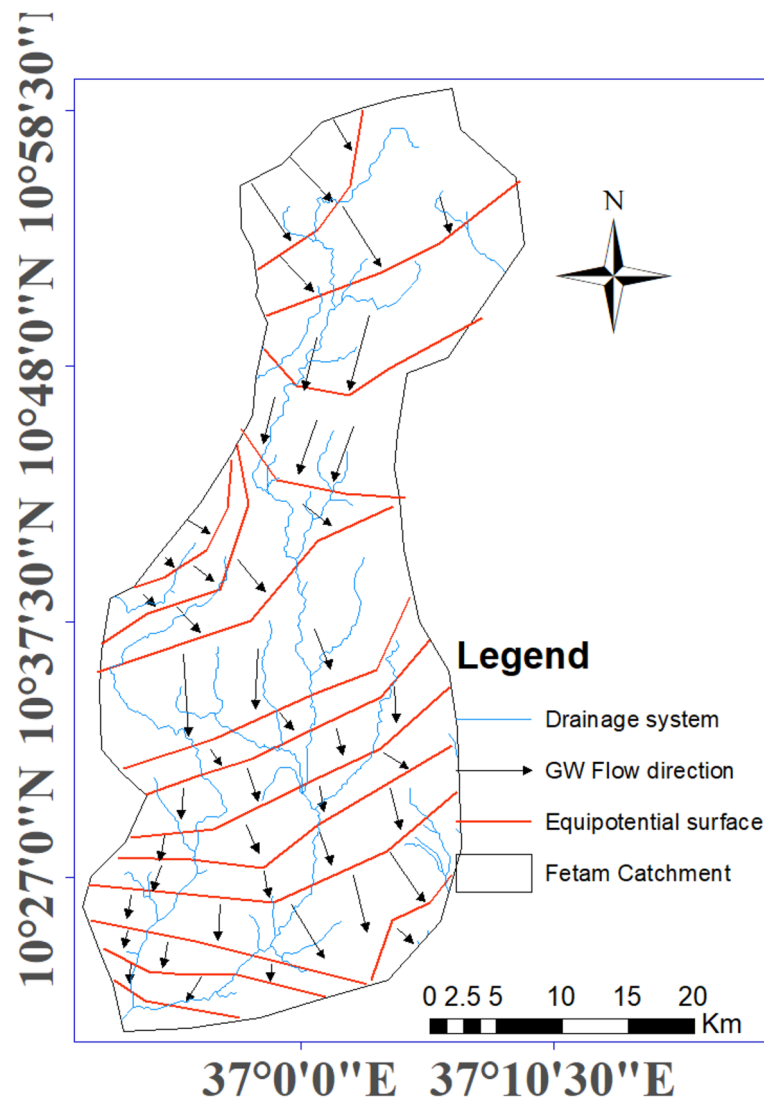


Fig. 10 Groundwater flow direction map

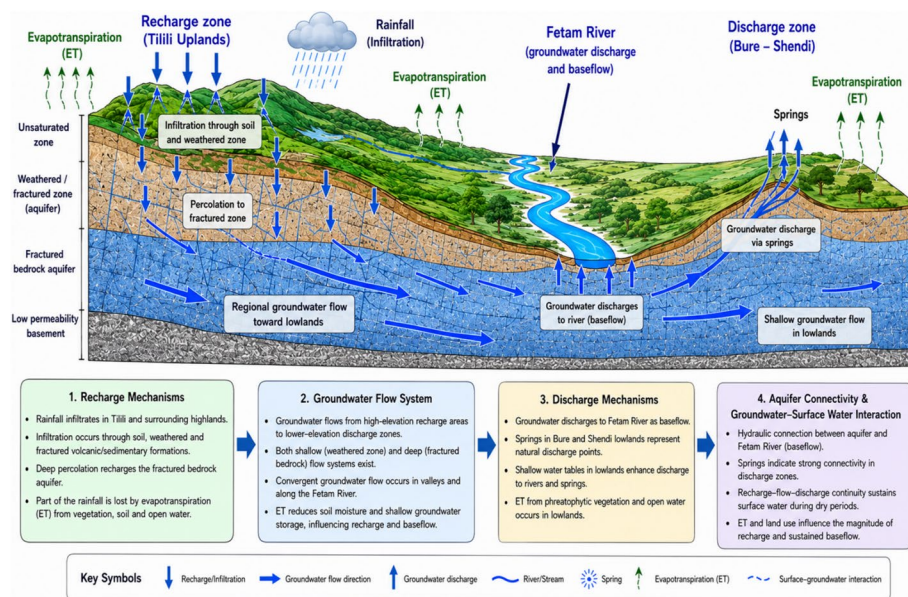


Fig. 11 Conceptual model of the groundwater system

discharge, and groundwater–surface water interaction processes, whereby rainfall in the Tilili uplands infiltrates through the soil and weathered formations, with part of the water lost through evapotranspiration and the remainder percolating into the fractured-bedrock aquifer; groundwater subsequently flows from the high-elevation recharge areas toward the lower-lying parts of the catchment through shallow weathered-zone and deeper fractured-bedrock flow systems, with convergent flow toward valleys and the Fetam River, where groundwater discharges as baseflow, while shallow groundwater contributes to river flow and emerges through springs in the Bure–Shendi discharge zone, demonstrating hydraulic connectivity between the aquifer, springs, and Fetam River and maintaining surface-water flow during dry periods.

3.6 Recharge and discharge area

3.6.1 Recharge area

The primary source of groundwater in the area is precipitation in the Highlands. In addition, the area's surrounding perennial rivers recharged local groundwater. Many of the hand-dug and shallow wells along these rivers are replenished by such conditions in the dry season. The corresponding Rivers recharge locally deposited alluvium along the Fetam River. The highly elevated areas, such as Kurb Mountain, Chitty, and locally developed ridges, are possible sources of recharge to the groundwater of the adjacent plateau and foothill aquifers. A deep water table and no springs characterize recharge areas in the highland of the catchment, which has the highest elevation. The groundwater contour lines in the figure above are closely spaced in the highland and widely spaced in the lowland. On the other hand, the closely spaced contour lines indicate the occurrence of the water table at a deeper depth. In contrast, the widely spaced contour lines indicate the discharge area, indicating the presence of the water table near the surface.

3.6.2 Groundwater discharge area

The discharge zone of the catchment is saturated in the southern portion of the area, which is characterized by flat lands. In this area, the appearance of spring and the occurrence of water near the surface. Many springs emerge from the weathered and fractured parts of topographic breaks, intermountain depressions, deep gorges, and high cliffs. It is observed that the discharge of this groundwater is due to differences in hydraulic conductivity resulting from lithological variation along the cliff, variation in the extent of weathering of the same rock, and variation in the degree and extent of fracturing. Generally, low-lying areas can be considered discharge areas. Most water chemistry data from drilling organizations indicate that higher TDS concentrations are found in the low-lying parts of the catchment, and the lowest TDS is found in the recharge area around Tillili. In addition, the DEM of the area is classified using Surfer software, as shown in Fig. 12.

3.7 . Daily soil moisture balance

Daily groundwater recharge estimated using the soil moisture balance (SMB) method (Table 4) shows clear spatial contrasts across the catchment when station values are

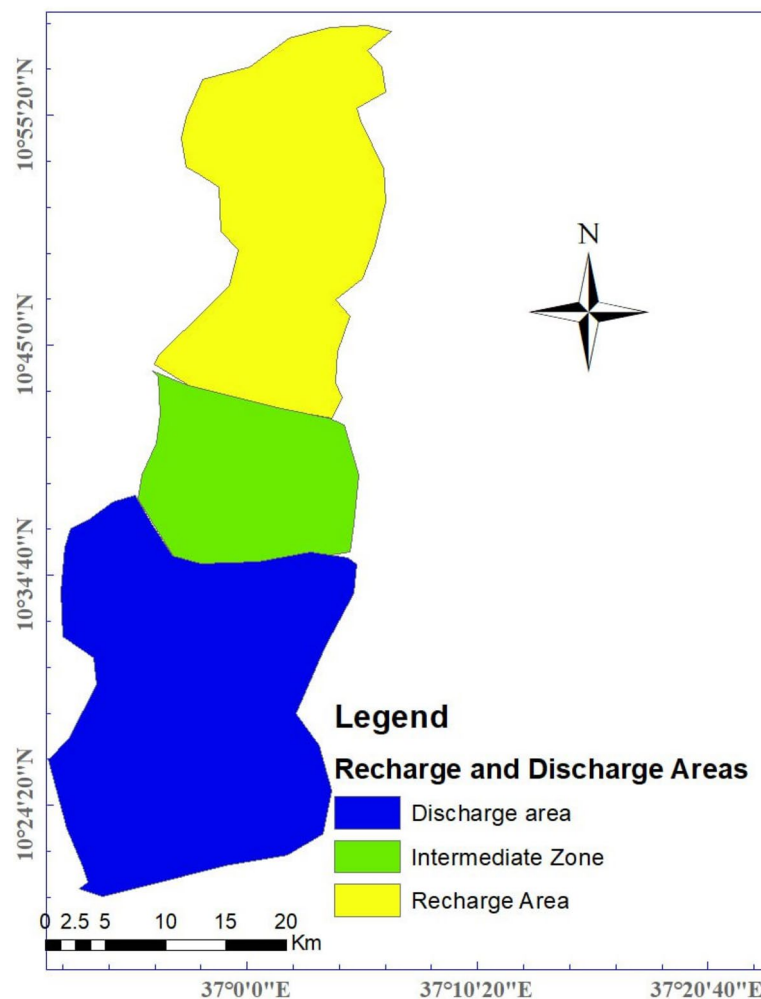


Fig. 12 Hydrogeologic zones

Table 4 Daily soil moisture balance results

Station	Precipitation	Area	Weighted-area	Weighted rain	Recharge	Weighted recharge
Tillili	2027	317	39.9	808.8	331.6	132.3
Bure	1524	103.9	13.1	199.6	23.5	3.08
Shendi	1739	373.8	47	817.3	75.5	35.5
Total		793.8	100	1825.7		170.5 ± 3.4

scaled by Thiessen polygon area weighting, a standard approach for converting point-based rainfall/recharge estimates into a catchment-average value [52]. The results indicate that Tillili has the highest recharge (331.6 mm/year) and also contributes the largest share to catchment recharge after weighting (132.3 mm/year) because it combines high rainfall (2027 mm) with a large contributing area (~39.9%). Shendi has moderate recharge (75.5 mm/year) and a weighted contribution of 35.5 mm/year, while Bure has the lowest recharge (23.5 mm/year) and contributes very little after weighting (3.08 mm/year) despite receiving substantial rainfall (1524 mm) and favorable geology, implying that local conditions strongly limit deep percolation there. Summed across all polygons, the total weighted recharge is 170.5 ± 3.41 mm/year with rainfall uncertainty, meaning that about 9.3% of the catchment's annual rainfall (based on the total weighted rainfall of 1825.7 mm) becomes groundwater recharge.

This partitioning is consistent with SMB theory, in which recharge occurs only after rainfall first satisfies soil moisture deficits and evapotranspiration demands; at finer (daily) time steps, the SMB method is better able to capture short wet-season recharge pulses than coarser aggregations [46, 52]. The recharge fractions further emphasize the dominance of the Tillili zone in the recharge budget: Tillili accounts for about 7.2% of catchment rainfall and 16.35% of its own weighted rainfall as recharge, whereas Shendi contributes 1.94% of catchment rainfall (about 4.3% of its weighted rainfall), and Bure contributes only 0.17% of catchment rainfall (about 1.54% of its weighted rainfall), indicating near-zero effective recharge in Bure relative to rainfall inputs summarized in Table 4. Runoff coefficients and PAW assumptions influence the estimated recharge rate by controlling the amount of precipitation available for potential groundwater recharge in the SMB method.

3.8 Monthly soil moisture balance

Monthly groundwater recharge estimated using the soil moisture balance (SMB) approach was computed for each rainfall–recharge zone defined by Thiessen polygons, allowing station-based recharge to be scaled to the catchment using area weighting [52]. The monthly SMB results show strong spatial contrasts: recharge at Tillili is highest (207.9 mm/year), followed by Shendi (65.6 mm/year), while Bure is lowest (11.5 mm/year). When these station estimates are multiplied by their polygon areas (39.6% for Tillili, 43.4% for Shendi, and 13% for Bure), the corresponding weighted contributions are 82.6, 28.5, and 1.5 mm/year, giving a catchment-total annual recharge of 112.7 ± 2.2 mm/year (Table 5). This indicates that most of the catchment's recharge contribution is controlled by the zones around Tillili and Shendi because they combine larger contributing areas with higher recharge rates. In contrast, Bure contributes little

Table 5 Monthly soil moisture balance

station	Area	area weighted area%	Excel sheet recharge	weighed recharge
Tillili	317.3	39.6	207.9	82.6
Bure	103.2	13	11.45	1.5
Shendi	373.3	43.4	65.6	28.5
Total	793.8	100	98.3	112.7 ± 2.2

due to both a smaller area fraction and a much lower recharge estimate. Using the Thiessen-weighted rainfall reference, the total monthly SMB recharge represents about 6.5% of annual rainfall, which is within the range commonly reported for recharge fractions derived from simplified water-balance/SMB approaches and reflects the fact that a large share of rainfall is lost to evapotranspiration and soil moisture replenishment before deep percolation becomes effective [46, 52, 64].

3.9 Base flow separation

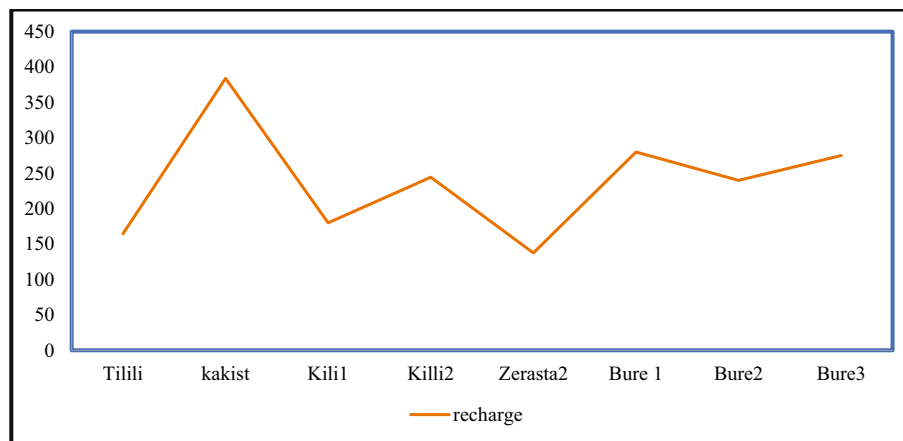
Baseflow separation indicates that streamflow in the catchment is composed of a large, quick-flow (direct runoff) component and a smaller, persistent baseflow component. The reported peak streamflow reaches 8.55 m³/s, of which 6.37 m³/s is attributed to direct runoff and 2.18 m³/s to baseflow. When these components are converted to catchment water depths, the annual runoff amounts to 712.4 mm/year, while the baseflow-derived recharge estimate is 244.9 ± 12.24 mm/year with errors derived from the measuring device. Interpreting recharge from baseflow is consistent with the widely used assumption that, over annual or multi-seasonal time scales, sustained baseflow largely represents delayed drainage from groundwater storage to the river network, making it a practical proxy for diffuse groundwater recharge in gauged basins [22, 49, 52]. Using the rainfall reference from the Thiessen polygon method, the recharge estimate corresponds to 14.2% of mean monthly rainfall, while the baseflow component represents 35.9% of the river's annual discharge, in contrast, total runoff accounts for 68.8% of annual discharge, emphasizing that storm runoff dominates the flow regime but that groundwater contributions remain substantial and hydrologically important [9, 49, 50]. Baseflow separation parameters, particularly the filter coefficient (α), number of filter passes, and initial baseflow conditions, affect the estimated recharge rate by controlling the separation of groundwater-derived baseflow from total streamflow.

3.10 Recession method

In the Fetam catchment, the hydrograph recession analysis identifies 101 recession periods, with the recession sequence beginning around the 344th day of the record when discharge is 1.59 m³/s and ending when discharge declines to 0.296 m³/s. Using recession-period accounting, the corresponding water volumes are reported as 623,529.4 m³ at the start and 1,392,564.7 m³ at the end of the recession, and the total baseflow contribution to the river during the recession is estimated at 6,230,964.7 m³. This type of analysis is consistent with standard baseflow/recession approaches, where the falling limb of streamflow is used to isolate the delayed groundwater contribution (baseflow) and quantify the groundwater-to-river flux over defined recession intervals [9, 49, 50, 52].

Table 6 Groundwater recharge by water level fluctuation

Hand-dug well stations	Easting	Northing	Elevation	Δh in mm	Existing soil or rocks	Sy	Recharge
TilliliZerasta1	283,994	1,200,904	2458	3500	Clay to silt clay	0.047	164.5 mm/year
Kakist	284,009	1,199,982	2424	4800	Silt	0.08	384 mm/year
Kili1	284,172	1,200,758	2450	6200	Clay	0.03	180 mm/year
Kili2	283,503	1,201,095	2448	8140	Clay	0.03	244.2 mm/year
Zerasta2	283,994	1,200,904	2458	2500	Silty clay	0.055	137.5 mm/year
Bure 1	289,757	1,184,232	2089	3500	Silt	0.08	280 mm/year
Bure2	286,922	1,182,435	2101	3000	Silt	0.03	90 mm/year
Bure3	287,027	1,182,464	2117	9160	Silty clay	0.03	275 mm/year
Ave. Recharge 220.1 ± 94.1 mm/year							

**Fig. 13** Groundwater recharge spatial variation

3.11 Water level fluctuation

The recharge estimated using the water-table fluctuation (WTF) method shows considerable spatial variability among the monitoring wells, ranging from 137.5 to 384 mm/year, with a mean of approximately 220.1 ± 94.1 mm/year (Table 6 and Fig. 13). This variability is mainly associated with differences in water-level fluctuation (Δh), specific yield (S_y), subsurface lithology, and local hydrogeological land use and hydrometeorological conditions (Yihdego, Y. & Webb, J., 2011). According to the WTF concept, recharge is proportional to water-table rise and specific yield; therefore, stations with larger Δh and/or higher S_y generally produce higher recharge estimates [21, 52]. Differences in silt, clay, and silty-clay materials can also influence infiltration and the timing of groundwater-level responses [16, 37, 45, 46]. Thus, the observed differences in recharge rates reflect the combined influence of geological, hydrogeological, and hydrometeorological factors across the catchment. The spatial variability of groundwater recharge among the monitoring stations was quantified using the standard deviation: $\sigma_R = \sqrt{\sum \left(\frac{R_i - R_{\text{mean}}}{n-1} \right)^2}$ where R_i is the recharge estimated at station i , R_{mean} is the mean recharge, and n is the total number of recharge-estimation stations. The calculated standard deviation was 94.1 mm/year, indicating substantial spatial variability in recharge among the stations. In the water-table fluctuation method, the estimated recharge is primarily influenced by

the magnitude of groundwater-level fluctuation Δh and the assigned specific yield S_y , such that $R = S_y \Delta h$; therefore, spatial variations in groundwater-level response and specific yield contribute to the observed variability in recharge estimates. The 94.1 mm/year represents the spatial variability (standard deviation) of the recharge estimates rather than a formal propagated measurement uncertainty.

3.12 Chloride mass balance

Groundwater recharge was also estimated using the chloride mass balance (CMB) approach, computing recharge separately with monthly and daily precipitation datasets and averaging the two estimates. Using the monthly dataset, the CMB-derived annual recharge is 210.9 mm/year, equivalent to 12.2% of the mean annual precipitation. Using the daily dataset, the estimated annual recharge is 226.1 mm/year, which corresponds to 12.3% of the annual rainfall. The final CMB recharge adopted for the catchment is the average of the two computations, giving 218.5 ± 8.2 mm/year resulted from a 4% uncertainty derived from measuring dive of chloride and rainfall recording gauges. The small difference between the monthly- and daily-based estimates reflects differences in precipitation time-step aggregation, while keeping the same conceptual mass-balance basis [15, 46, 52]. In the study area, the chloride concentrations in precipitation and runoff are reported to be very similar, with precipitation slightly higher than runoff, which increases the net chloride input term and results in a relatively higher recharge estimate compared to situations where runoff chloride is higher [1, 10, 11, 64]. The estimated recharge using the chloride mass balance (CMB) method is influenced by spatial and temporal variability in chloride concentrations in precipitation and groundwater, as these variations directly affect the calculated recharge rate.

3.13 Water balance

The water-balance method estimated groundwater recharge at 187 ± 3.7 mm/year using the Thornthwaite–Mather approach and 143 ± 2.8 mm/year using the Excel-based water-balance calculation, corresponding to 13.6% and 5.2% of the mean annual precipitation, respectively. The relatively low recharge estimates are partly related to the structure of the water-balance calculation, in which the available water is preferentially allocated to satisfy actual evapotranspiration (AET), thereby constraining the residual water available for recharge, which is the main component for groundwater recharge assessment. The 146 mm/year difference between the Thornthwaite–Mather (956 mm/year) and Penman–Monteith (810 mm/year) AET estimates reflects methodological differences and underlying assumptions, including the use of temperature-based empirical relationships in Thornthwaite–Mather versus a physically based approach incorporating radiation, air temperature, humidity, and wind speed in Penman–Monteith this my makes different recharge uncertainty in water balnce approach and take average of the two is more preferable.

The difference should be interpreted as an expression of methodological uncertainty in AET estimation and its subsequent influence on the water balance and recharge. An additional source of uncertainty is the hydrogeological boundary condition of the Fetam Catchment. The catchment contains an interconnected weathered and fractured aquifer system that can transmit groundwater across the surface-water divide; therefore, the

system should be considered partly open from a groundwater-flow perspective. Consequently, the water-balance approach may not explicitly account for lateral groundwater inflow or outflow across the catchment boundary. If groundwater leaves the catchment through the fractured aquifer, the calculated residual recharge may be underestimated, whereas groundwater inflow from adjacent areas may produce an apparent overestimation of local recharge. Thus, the estimated 143–187 mm/year should be regarded as a catchment-scale recharge estimate with uncertainty associated with groundwater boundary fluxes and seasonal water-storage fluctuations. Nevertheless, the magnitude of recharge is considered reasonable because it is broadly consistent with the independent recharge estimates obtained from the other methods applied in this study.

4 Discussion

Recharge in the Fetam catchment was estimated using five independent methods to reduce method-specific bias and to constrain uncertainty by combining evidence from the unsaturated zone, saturated zone, tracer mass balance, streamflow response (baseflow separation), and basin-scale accounting (water balance). This approach is consistent with watershed-scale studies showing that convergence across methods provides a more defensible basis for management than reliance on a single estimator [27, 35, 51]. Across the applied methods, annual recharge ranges from 112.7 to 244.9 mm/year, indicating a moderate spread among estimators within the same hydroclimatic setting. The largest estimates were obtained from baseflow separation (244.9 mm/year; 10.7% of annual rainfall) and water table fluctuation (220.1 mm/year; 13.8%), followed by chloride mass balance (218.6 mm/year; 12.3%). The lower estimates were produced by daily SMB (170.5 mm/year; 9.3%), water balance (165 mm/year; 9.5%), and monthly SMB (112.7 mm/year; 6.5%).

Given the strong rainfall seasonality and the dry-season dependence on groundwater, defining a realistic recharge range is essential for evaluating whether abstraction can be sustained without progressive water-level decline and loss of dry-season baseflow support. Baseflow separation produced the highest recharge estimate [9, 19].

The relatively higher recharge estimates obtained from the WTF and baseflow-separation methods should be interpreted as potential upper-bound estimates because both methods may capture water contributions beyond direct diffuse aquifer recharge. In the WTF method, observed groundwater-level fluctuations may reflect delayed interflow, seasonal release of shallow subsurface storage, and localized focused recharge, whereas hydrograph separation may attribute contributions from bank storage, wetlands, temporary saturated zones, and shallow groundwater responses to baseflow. These processes can increase the estimated recharge when the separated baseflow is assumed to represent groundwater discharge from the aquifer. Furthermore, recharge in the Fetam Catchment may occur through both diffuse recharge, resulting from relatively widespread infiltration of precipitation through the soil and unsaturated zone, and focused recharge, where water is concentrated and rapidly infiltrates through stream channels, topographic depressions, fractures, and other preferential pathways. Therefore, the recharge estimates from WTF and baseflow separation represent integrated hydrological responses and may include both diffuse and focused recharge components, as well as delayed subsurface storage contributions. This distinction is important when comparing

recharge estimates obtained from different methods and interpreting their associated uncertainty [10, 50].

The WTF method yielded a recharge estimate close to the baseflow result. The study interpretation is that WTF is a strong option here because it is based on direct groundwater-level observations and is less affected by evapotranspiration once recharge reaches the aquifer [23, 44]. Seasonal groundwater-level rise during the rainy season and decline during the dry season create a pronounced amplitude interpreted as recharge input to storage [29, 45]. The study also notes rapid aquifer response around Tillili, supporting the plausibility of strong seasonal recharge signals in shallow groundwater [37]. CMB produced an intermediate-high recharge estimate (~ 218.5 mm/year), which is reasonable as it reflects longer-term effective recharge integrated through solute concentration rather than event-scale soil accounting [6, 14].

In this catchment, the chloride concentrations in precipitation and runoff are reported to be similar, but slightly higher in precipitation, which increases net chloride input and yields a relatively higher recharge estimate than would occur if runoff chloride were higher. The reliability of the CMB estimate depends strongly on assumptions of conservative chloride behavior and limited additional chloride sources, and the study notes that limited temporal sampling constrains confidence in year-to-year representativeness [1, 38]. The daily SMB estimate (170.5 mm/year) is lower than WTF/baseflow/CMB, as expected, because SMB explicitly accounts for large evaporative losses and runoff before deep percolation [11, 20]. The study highlights that the SMB implementation is highly sensitive to key inputs—especially runoff coefficient, initial soil moisture, and PAW—and that the catchment's combination of cultivated land, flat terrain, relatively high runoff coefficient, and high AET tends to reduce modeled recharge [65].

The monthly SMB estimate is the lowest among all methods [2, 52]. This is attributed in the study to differences introduced by time-step aggregation and lower annual precipitation totals when using monthly data compared to daily accounting—effects that can dampen short wet-season pulses responsible for episodic drainage in clay-rich soils [15, 46, 52]. The water balance method yielded 165 mm/year, broadly consistent with the daily SMB magnitude [2, 12]. However, the study emphasizes that the catchment behaves as an open groundwater system, implying potential subsurface inflow/outflow across boundaries in deeper fractured aquifers [16, 21]. Because these fluxes are not quantified, the method assumes groundwater inflow equals groundwater outflow, which can bias recharge either upward or downward depending on the true net boundary exchange [11, 16, 52].

The recharge pattern is consistent with the catchment's groundwater flow system and physiography [16]. Groundwater generally flows from higher-head recharge areas toward lower-head discharge areas [16], and the study identifies Tillili and the surrounding highlands as major recharge areas, while parts of Bure and Shendi function as discharge areas with relatively smaller annual recharge. Mapped groundwater flow directions and the association of discharge areas with shallow water tables and spring occurrence support this conceptualization, whereas recharge areas show deeper water tables and fewer springs [16, 21]. Seasonality also strongly controls recharge because the wet season produces the rainfall pulses needed to overcome soil moisture deficit and initiate deep percolation, while the dry season is dominated by soil moisture depletion

and groundwater recession [21, 37, 45, 46]. AET peaks during the late dry season (April–May), and PET and AET tend to converge during the wet season when water availability is high—conditions that influence the timing of effective recharge and the partitioning of rainfall losses [12, 31, 68].

From a management perspective, the multi-method results define a realistic recharge envelope for planning abstraction in a catchment where groundwater is a primary dry-season buffer. The higher estimates (baseflow/WTF) likely represent upper bounds influenced by streamflow separation behavior and strong seasonal water-level rise, while SMB and water balance provide lower-to-mid estimates that explicitly reflect evaporative demand and runoff losses. Key uncertainties arise from: (i) limited temporal groundwater-level monitoring for WTF, (ii) limited temporal chloride sampling for CMB, (iii) sensitivity of SMB to runoff coefficients and PAW in clay-rich soils, and (iv) unquantified groundwater boundary fluxes for water balance in an open fractured-aquifer system.

4.1 Groundwater management implications

The estimated groundwater recharge provides an important basis for evaluating the groundwater development potential of the Fetam Catchment. Based on the estimated annual groundwater recharge of 149.71 billion L/year, a preliminary groundwater allocation framework was established to balance groundwater development with the maintenance of natural groundwater-system functions. In this framework, 60% of the total recharge, equivalent to 89.83 billion L/year, was allocated to natural discharge and environmental sustainability, including groundwater storage, spring discharge, stream baseflow, wetlands, and groundwater-dependent ecosystems. The remaining 40% of the total recharge, equivalent to 59.88 billion L/year, was considered potentially available for abstraction.

The potentially abstractable groundwater was further allocated among the major water-use sectors. Of the 59.88 billion L/year potentially available for abstraction, 70% (41.92 billion L/year) was allocated to irrigation, 23% (13.77 billion L/year) to domestic water supply, and 7% (4.19 billion L/year) to industrial water use. This framework provides a preliminary water-allocation scenario rather than a fixed estimate of sustainable yield, because the actual amount of groundwater that can be abstracted depends on aquifer properties, spatial and temporal recharge variability, groundwater storage, natural discharge, and groundwater–surface-water interactions.

Under the current population scenario, 250,000 households with an average household size of five persons correspond to a total population of 1.25 million people. A domestic water requirement of 25 L/capita/day, the current annual domestic water demand is estimated to be 11.41 billion L/year. This demand represents approximately 82.8% of the preliminary domestic allocation of 13.77 billion L/year, leaving approximately 2.36 billion L/year within the allocated domestic-water component.

Considering population growth, an annual growth rate of 2.6% was applied to project the population over a 10-year period. The projected population increases from approximately 1.25 million to 1.62 million people, resulting in an estimated domestic water demand of approximately 14.75 billion L/year, and the water requirement remains constant at 25 L/capita/day. This projected demand would exceed the current domestic allocation of 13.77 billion L/year by approximately 0.98 billion L/year. The result indicates

that population growth alone could place increasing pressure on the domestic groundwater allocation during the planning period.

The irrigation allocation of approximately 41.92 billion L/year and industrial allocation of approximately 4.19 billion L/year represent preliminary planning limits for these sectors. However, the actual abstraction should be controlled according to groundwater-level trends, well yields, aquifer storage and transmissivity, spring discharge, streamflow, and other indicators of groundwater-system response. In addition, climate change and land-use/land-cover (LULC) change may alter groundwater recharge and water demand through changes in precipitation, evapotranspiration, infiltration, runoff, vegetation cover, agricultural expansion, and urbanization. Therefore, future groundwater-resource management should incorporate population-growth, climate-change, and LULC scenarios to reassess recharge, natural discharge requirements, and sectoral abstraction limits. The proposed 60% allocation for natural discharge and environmental sustainability and 40% for abstraction, divided into 70% irrigation, 23% domestic, and 7% industrial use, should consequently be regarded as a preliminary precautionary management scenario that can be refined through long-term groundwater monitoring and integrated groundwater-flow and recharge modelling. Future hydrogeological work should include Numerical groundwater flow modelling, Isotope-based recharge assessment, Long-term groundwater hydrograph analyses, and integrated groundwater–surface-water modelling.

5 Conclusions

Groundwater recharge in the Fetam Catchment is strongly seasonal and spatially heterogeneous. Five independent methods produced annual recharge estimates of 113–245 mm/year, equivalent to approximately 6.5–14% of mean annual rainfall. Baseflow separation and water-table fluctuation yielded the highest values and are interpreted as possible upper bounds because they may include delayed interflow, bank storage, shallow groundwater responses, and focused recharge. Soil-moisture balance and catchment water-balance methods produced lower estimates because they explicitly account for evapotranspiration and runoff losses, while chloride mass balance provided an intermediate estimate representative of longer-term effective recharge. The combined evidence supports a defensible management range of approximately 165–220 mm/year. Recharge is concentrated in the Tillili uplands and other higher-elevation fractured and permeable zones, whereas lower areas around Bure and Shendi mainly function as discharge zones connected to springs and river baseflow. The multi-method comparison demonstrates that recharge should be managed as a range rather than as a single deterministic value. Groundwater development should therefore be linked to continued monitoring of groundwater levels, spring discharge, streamflow, and hydrochemistry, with abstraction adjusted as new data refine recharge estimates and aquifer-response thresholds.

Author contributions

Mekonen Gedewon Aynalem: Writing – original draft, Visualization, Methodology, Investigation, Data curation, Conceptualization, Writing – reviewing and editing. Sileshi Mamo: Writing – original draft, Visualization, Methodology, Investigation, Data curation, Conceptualization, Writing – reviewing and editing. Melkamu Meseret: Writing – original draft, Visualization, Methodology, Investigation, Data curation, Conceptualization, Writing – reviewing and editing. Muralitharan Jothamani: Writing – original draft, Visualization, Methodology, Investigation, Data curation, Conceptualization,

Writing – reviewing and editing. Habtamu Wondem: Writing – original draft, Visualization, Methodology, Investigation, Data curation, Conceptualization, Writing – reviewing and editing.

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Data availability

The remote sensing data used in this study were obtained from the publicly available Copernicus Sentinel-2 satellite mission through the Copernicus Data Space Ecosystem (<https://dataspace.copernicus.eu/>). The meteorological, hydrological, groundwater-level, hydrochemical, soil, land use/land cover, and hydrogeological datasets used in this study were collected from relevant governmental agencies, field measurements, and laboratory investigations. These datasets are not publicly available due to institutional and field-data restrictions but can be obtained from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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References

- Alcalá F, Custodio E. Natural uncertainty of spatial average aquifer recharge through atmospheric chloride mass balance in continental Spain. *J Hydrol.* 2015;524:642–61. <https://doi.org/10.1016/j.jhydrol.2015.03.018>.
- Andualem T, Demeke G, Ahmed I, Dar M, Yibeltal M. Groundwater recharge estimation using empirical methods from rainfall and streamflow records. *J Hydrol Reg Stud.* 2021. <https://doi.org/10.1016/j.ejrh.2021.100917>.
- Ardekani A, Sabzevari T, Haghighi A, Petroselli A. Separation of surface flow from subsurface flow in catchments using runoff coefficient. *Acta Geophys.* 2021;69:2363–76. <https://doi.org/10.1007/s11600-021-00667-6>.
- Aynalem MG, Fentaye SM, Alemu MM, Gizaw EA. Groundwater recharge assessment and hydrometeorological scenario in the Fetam catchment Upper Blue Nile Basin, Ethiopia. 2024. <https://doi.org/10.22541/essoar.171805143.35325282/v1>
- Bazuhaire AS, Wood WW. Chloride mass-balance method for estimating ground water recharge in arid areas: examples from western Saudi Arabia. *Journal of Hydrology.* 1996;186(1–4): 153–159. [https://doi.org/10.1016/s0022-1694\(96\)03028-4](https://doi.org/10.1016/s0022-1694(96)03028-4)
- Bekele S, Geremew G, Ayele E. Estimation of Annual and Seasonal Groundwater Recharge by Using Wetpass-M and Chloride Mass-Balance Methods, Upper Bilate Catchment, Rift Valley Basin Ethiopia. *Adv Civil Eng.* 2024. <https://doi.org/10.1155/2024/2724015>.
- Belachew K, Maina N, Dersseh W, Zeleke B, Stoddard F. Yield gaps of major cereal and grain legume crops in Ethiopia: a review. *Agronomy.* 2022. <https://doi.org/10.3390/agronomy12102528>.
- Bennett G, Shemsanga C, Kervyn M, Walraevens K. Groundwater recharge estimation using groundwater level fluctuations and baseflows around Mount Meru, Tanzania. *Groundw Sustain Dev.* 2024. <https://doi.org/10.1016/j.gsd.2024.101133>.
- Bhardwaj S, Jha M, Uniyal B. Assessing efficacy of baseflow separation techniques in a Himalayan River Basin, Northern India. *Environ Process.* 2024. <https://doi.org/10.1007/s40710-024-00680-z>.
- Crosbie R, Rachakonda P. Constraining probabilistic chloride mass-balance recharge estimates using baseflow and remotely sensed evapotranspiration: the Cambrian Limestone Aquifer in northern Australia. *Hydrogeol J.* 2021;29:1399–419. <https://doi.org/10.1007/s10040-021-02323-1>.
- Crosbie R, Peeters L, Herron N, McVicar T, Herr A. Estimating groundwater recharge and its associated uncertainty: use of regression kriging and the chloride mass balance method. *J Hydrol.* 2017. <https://doi.org/10.1016/j.jhydrol.2017.08.003>.
- Da Costa Barros F. Thornthwaite and mather soil water balance model adapted for estimation of real evapotranspiration of the pasture. *Revista Engenharia na Agricultura - Reveng.* 2021. <https://doi.org/10.13083/reveng.v29i1.11703>.
- Danebo D, Atilebachew A, Abebe A, Jothamani M. Applications of Geospatial Technologies and Wetpass Model in Groundwater Recharge Estimation in Sana River Catchment, Kembata Tembaro Zone, Southern Ethiopia. In: Nandagiri, L., Narasimhan, M.C., Marathe, S. (eds) Recent Advances in Civil Engineering. CTCS 2021. Lecture Notes in Civil Engineering, vol 256. Springer, Singapore. 2023. https://doi.org/10.1007/978-981-19-1862-9_6
- Demlie M. Assessment and estimation of groundwater recharge for a catchment located in highland tropical climate in central Ethiopia using catchment soil–water balance (SWB) and chloride mass balance (CMB) techniques. *Environ Earth Sci.* 2015;74:1137–50. <https://doi.org/10.1007/s12665-015-4099-y>.

15. Dourado-Neto D, Lier Q, Metselaar K, Reichardt K, Nielsen D. General procedure to initialize the cyclic soil water balance by the Thornthwaite and Mather method. *Sci Agric*. 2010;67:87–95. <https://doi.org/10.1590/s0103-90162010000100013>.
16. Freeze RA, Cherry JA. *Groundwater* prentice-hall. Englewood Cliffs, NJ. 1979;176:161–77.
17. Gebre T, Kibru T, Tesfaye S, Taye G. Analysis of watershed attributes for water resources management using GIS: the case of Chelekot Micro-Watershed, Tigray, Ethiopia. *J Geogr Inf Syst*. 2015;07:177–90. <https://doi.org/10.4236/jgis.2015.72015>.
18. Gebru A, Gebreyohannes T, Kahsay G, Grum B. The dilemma of choosing appropriate groundwater recharge estimation methods in Ethiopia: a systematic review of the existing methods. *Groundw Sustain Dev*. 2024. <https://doi.org/10.1016/j.gsd.2024.101358>.
19. Gidafie D, Nedaw D, Azagegn T, Abebe B, Baba A. Estimating groundwater recharge through multiple methods: southern sections of the western Afar rift margin and associated rift floor. *Environ Earth Sci*. 2024. <https://doi.org/10.1007/s12665-024-11999-z>.
20. Gumula-Kawecka A, Jaworska-Szulc B, Szymkiewicz A, Gorczewska-Langner W, Pruszkowska-Caceres M, Angulo-Jaramillo R, et al. Estimation of groundwater recharge in a shallow sandy aquifer using unsaturated zone modeling and water table fluctuation method. *J Hydrol*. 2021. <https://doi.org/10.1016/j.jhydrol.2021.127283>.
21. Healy RW, Cook PG. Using groundwater levels to estimate recharge. *Hydrogeol J*. 2002;10:91–109. <https://doi.org/10.1007/s10040-001-0178-0>.
22. Huang Z, Yeh P, Jiao J, Luo X, Pan Y, Long Y, et al. A new approach for assessing groundwater recharge by combining GRACE and baseflow with case studies in Karst areas of Southwest China. *Water Resour Res*. 2023. <https://doi.org/10.1029/2022wr032091>.
23. Jia W, Yin L, Zhang M, Zhang X, Zhang J, Tang X, et al. Quantification of groundwater recharge and evapotranspiration along a semi-arid wetland transect using diurnal water table fluctuations. *J Arid Land*. 2021;13:455–69. <https://doi.org/10.1007/s40333-021-0100-7>.
24. Jiang A, Liu Y, Yi S, Guo Z, Cui Y, Huang Q, et al. Impact of artificial recharge on groundwater and springs: Jinan, China, case study. *Arab J Geosci*. 2021;14:1–10. <https://doi.org/10.1007/s12517-020-06394-4>.
25. Jiang S, Rao W, Han L, Meredith K. Study of groundwater recharge using combined unsaturated-and saturated-zone chloride mass balance methods. *Hydrol Process*. 2023. <https://doi.org/10.1002/hyp.14927>.
26. Kadirkhodjaev A, et al. Assessing climate change impacts on groundwater recharge and storage using MODFLOW in the Akhangaran River Alluvial Aquifer, Eastern Uzbekistan. *Water*. 2025. <https://doi.org/10.3390/w17152291>.
27. Koeniger P, Gaj M, Beyer M, Himmelsbach T. Review on soil water isotope-based groundwater recharge estimations. *Hydrol Process*. 2016;30:2817–34. <https://doi.org/10.1002/hyp.10775>.
28. Konapala G, Mishra A, Wada Y, Mann M. Climate change will affect global water availability through compounding changes in seasonal precipitation and evaporation. *Nat Commun*. 2020. <https://doi.org/10.1038/s41467-020-16757-w>.
29. Kourakos G, Dahlke H, Harter T. Increasing groundwater availability and seasonal base flow through agricultural managed aquifer recharge in an irrigated basin. *Water Resour Res*. 2019;55:7464–92. <https://doi.org/10.1029/2018wr024019>.
30. Kourouma J, Eze E, Negash E, Phiri D, Vinya R, Girma A, et al. Assessing the spatio-temporal variability of NDVI and VCI as indices of crops productivity in Ethiopia: a remote sensing approach. *Geomat Nat Hazards Risk*. 2021;12:2880–903. <https://doi.org/10.1080/19475705.2021.1976849>.
31. Lai Y, Tian J, Kang W, Guo S, Zhou Y, He C. Estimating evapotranspiration from soil moisture using the improved soil water balance method in cold mountainous areas. *J Hydrol X*. 2023. <https://doi.org/10.1016/j.hydroa.2023.100154>.
32. Laoufi A, Guettaia S, Boudjema A, Derdour A, Almalki A, Bojer A, et al. Seasonal groundwater quality analysis in a drought prone agricultural region using GIS and IWQI for nitrate contamination insights. *Sci Rep*. 2025. <https://doi.org/10.1038/s41598-025-06884-z>.
33. Leenaars J, Claessens L, Heuvelink GBM, Hengl T, González MR, Van Bussel LGJ, et al. Mapping rootable depth and root zone plant-available water holding capacity of the soil of sub-Saharan Africa. *Geoderma*. 2018;324:18–36. <https://doi.org/10.1016/j.geoderma.2018.02.046>.
34. Lemann T, Roth V, Zeleke G, Subhatu A, Kassawmar T, Hurni H. Spatial and temporal variability in hydrological responses of the Upper Blue Nile basin, Ethiopia. *Water*. 2018. <https://doi.org/10.3390/w11010021>.
35. Li Z, Chen X, Liu W, Si B. Determination of groundwater recharge mechanism in the deep loessial unsaturated zone by environmental tracers. *Sci Total Environ*. 2017;586:827–35. <https://doi.org/10.1016/j.scitotenv.2017.02.061>.
36. Lindle J, Villholth K, Ebrahim G, Sorensen J, Taylor R, Jensen K. Groundwater recharge influenced by ephemeral river flow and land use in the semiarid Limpopo Province of South Africa. *Hydrogeol J*. 2023;31:2291–306. <https://doi.org/10.1007/s10040-023-02682-x>.
37. Liu H, Tang J, Zhang X, Wang R, Zhu B, Li N, et al. Seasonal variations of groundwater recharge in a small subtropical agroforestry watershed with horizontal sedimentary bedrock. *J Hydrol*. 2020. <https://doi.org/10.1016/j.jhydrol.2020.125703>.
38. Liu J, Chen Z, Nie Z, Tian Y. Estimation of groundwater recharge from precipitation by using tritium and chloride concentrations in the Chinese loess unsaturated zone in Inner Mongolia, China. *Hydrogeol J*. 2024;32:1511–20. <https://doi.org/10.1007/s10040-024-02798-8>.
39. Mahmoud H, Mahmood T, Korom S. Groundwater recharge dynamics to recent wetting in a cold region aquifer. *J Hydrol Reg Stud*. 2025. <https://doi.org/10.1016/j.ejrh.2025.102902>.
40. Mekonnen Z, Woldeamanuel T, Kassa H. Socio-ecological vulnerability to climate change/variability in central rift valley, Ethiopia. *Adv Clim Change Res*. 2019. <https://doi.org/10.1016/j.jaccr.2019.03.002>.
41. Mengiste W, Tsegaye D, Dagalo S, Yitbareke T. Characterization and classification of soils in the irrigable land of Achewa small-scale irrigation area, Itang special district, southwest Gambella, Ethiopia. *Front Soil Sci*. 2025. <https://doi.org/10.3389/fsoil.2024.1451271>.

42. Nyssen J, Tielsens S, Gebreyohannes T, Araya T, Teka K, Van De Wauw J, et al. Understanding spatial patterns of soils for sustainable agriculture in northern Ethiopia's tropical mountains. *PLoS ONE*. 2019. <https://doi.org/10.1371/journal.pone.0224041>.
43. Oyda Y, Hatiye SD, Jothimani M. GIS-based WetSpa-M modeling for water balance components estimation considering geological conditions in data-limited regions: case of the Maze-Zenti catchment, Ethiopia. *J Afr Earth Sci*. 2026;239:106133. <https://doi.org/10.1016/j.jafrearsci.2026.106133>.
44. Penner G, Martins R, Rodrigues S, Wendland E. Uncertainty in groundwater recharge estimation using groundwater level fluctuation and aquifer test. *RBRH*. 2023. <https://doi.org/10.1590/2318-0331.282320220113>.
45. Priestley S, Baker A, Shanafield M, Timms W, Andersen M, et al. Groundwater recharge of fractured rock aquifers in SE Australia is episodic and controlled by season and rainfall amount. *Geophys Res Lett*. 2025. <https://doi.org/10.1029/2024gl113503>.
46. Rushton K, Eilers V, Carter R. Improved soil moisture balance methodology for recharge estimation. *J Hydrol*. 2006;318:379–99. <https://doi.org/10.1016/j.jhydrol.2005.06.022>.
47. Saedi F, Kumar M, Clement T. Development of a process-based modeling framework for harnessing hydrometeorological datasets to estimate groundwater recharge. *J Environ Manage*. 2025;384:125470. <https://doi.org/10.1016/j.jenvman.2025.125470>.
48. Santos R, Koide S, Távora B, De Araujo D. Groundwater recharge in the Cerrado Biome, Brazil—a multi-method study at experimental watershed scale. *Water*. 2020. <https://doi.org/10.3390/w13010020>.
49. Schilling K, Langel R, Wolter C, Arenas-Amado A. Using baseflow to quantify diffuse groundwater recharge and drought at a regional scale. *J Hydrol*. 2021;602:126765. <https://doi.org/10.1016/j.jhydrol.2021.126765>.
50. Shao G, Zhang D, Guan Y, Sadat M, Huang F. Application of different separation methods to investigate the baseflow characteristics of a Semi-Arid Sandy Area, Northwestern China. *Water*. 2020. <https://doi.org/10.3390/w12020434>.
51. Shi P, Huang Y, Yang C, Li Z. Quantitative estimation of groundwater recharge in the thick loess deposits using multiple environmental tracers and methods. *J Hydrol*. 2021. <https://doi.org/10.1016/j.jhydrol.2021.126895>.
52. Singh A, Panda S, Uzokwe V, Krause P. An assessment of groundwater recharge estimation techniques for sustainable resource management. *Groundw Sustain Dev*. 2019. <https://doi.org/10.1016/j.gsd.2019.100218>.
53. Sun X, Zhang B, Dai M, Gao R, Jing C, Gu S, et al. Research on methods for estimating reference crop evapotranspiration under incomplete meteorological indicators. *Front Plant Sci*. 2024;15:1354913. <https://doi.org/10.3389/fpls.2024.1354913>.
54. SUTCLIFFE DW, CARRICK TR, HERON J, RIGG E, TALLING JF, WOOF C, LUND JW. Long-term and seasonal changes in the chemical composition of precipitation and surface waters of lakes and tarns in the English Lake District. *Freshwater Biology*. 1982;12(5):451–506. Portico. <https://doi.org/10.1111/j.1365-2427.1982.tb00640.x>.
55. Tadele T, Degu Y, Mechal A. Land use and land cover dynamics of the Weito River catchment in the Ethiopian rift using remote sensing and CA Markov modelling approaches. *Discov Sustain*. 2025. <https://doi.org/10.1007/s43621-025-01670-3>.
56. Tadesse G, Jothimani M. Assessing groundwater recharge in the Wabe River catchment, Central Ethiopia, through a GIS-based distributed water balance model. *Earth*. 2024;5(1):20–44. <https://doi.org/10.3390/earth5010002>.
57. Takele T, Bayisa A, Jothimani M. Impacts of land use and land cover changes on groundwater recharge in the Dire Dawa watershed Ethiopia. *Asian J Water Environ Pollut*. 2025;22(6):103. <https://doi.org/10.36922/ajwep025180139>.
58. Thornthwaite CW, Mather JR. The water balance. Centerton, NJ: Drexel Institute of Technology-Laboratory of Climatology. 104p. Publications in Climatology. 1955;8(1).
59. Tufa F, Feyissa F, Kebede A, Gudeta B, Kiteessa W, Debela S, et al. Estimation of groundwater recharge in a volcanic aquifer system using soil moisture balance and baseflow separation methods: the case of Gilgel Gibe Catchment, Ethiopia. *Hydrology*. 2024. <https://doi.org/10.3390/hydrology11070109>.
60. Wakjira M, Peleg N, Six J, Molnar P. Current and future cropland suitability for cereal production across the rainfed agricultural landscapes of Ethiopia. *Agric For Meteorol*. 2024. <https://doi.org/10.1016/j.agrformet.2024.110262>.
61. Walczuk A, Campos J, Teles L. Shallow groundwater and river baseflow: interconnection evidences and integrated water management strategies. Case study in the Upper Descoberto Watershed, Federal District, Brazil. *Journal of South American Earth Sciences*. 2024. <https://doi.org/10.1016/j.jsames.2024.105043>.
62. Woldu T, Ayenew T, Baychken B, Birhanu B. Spatiotemporal recharge estimation in the upper Awash sub-basin, central Ethiopia. *Hydrol Res*. 2024. <https://doi.org/10.2166/nh.2024.164>.
63. Wood WW, Sanford WE. Chemical and Isotopic Methods for Quantifying Ground-Water Recharge in a Regional, Semiarid Environment. *Groundwater* 1995;33(3):458–468. Portico. <https://doi.org/10.1111/j.1745-6584.1995.tb00302.x>.
64. Xu Y, Beekman HE. Groundwater Recharge Estimation in Southern Africa. UNESCO IHP Series No. 64. 2003.
65. Yenehun A, Nigate F, Belay A, Desta M, Van Camp M, Walraevens K. Groundwater recharge and water table response to changing conditions for aquifers at different physiography: The case of a semi-humid river catchment, north-western highlands of Ethiopia. *Sci Total Environ*. 2020;748:142243. <https://doi.org/10.1016/j.scitotenv.2020.142243>.
66. Yihdego Y, Webb J. An empirical water budget model as a tool to identify the impact of land-use change in stream flow in southeastern Australia. *Water Resour Manag*. 2013;27(14):4941–58.
67. Ying Z, Tetzlaff D, Comte J, Wu S, Soulsby C. Storage dynamics and groundwater-surface water interactions in a drought-sensitive lowland catchment: process-based modelling as a learning tool. *Hydrol Process*. 2025. <https://doi.org/10.1002/hyp.70141>.
68. Zotarelli L, Dukes MD, Romero CC, Migliaccio KW, Morgan KT. Step-by-step calculation of the Penman-Monteith evapotranspiration (FAO-56 method). *EDIS*. 2024. <https://doi.org/10.32473/edis-ae459-2010>.

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